Russia: National Innovation Strategy Brief

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Country assessment: Russia's national innovation system

Russia has strong scientific talent but turns relatively little of it into innovation. In WIPO's Global Innovation Index (GII) 2025, Russia ranks 60th, far behind the United States (3rd), China (10th), and France (13th), and it has dropped from 45th in 2021 [2]. Its best area is human capital and research (28th). Its worst is institutions (131st), by far the lowest of the four countries in the game. Russia does rank higher on innovation outputs (55th) than on inputs (73rd), so it gets more out of its resources than its spending would suggest [2].

Indicator	Russia	United States	China	France
GII 2025 overall rank [2]	60	3	10	13
Human capital and research, rank [2]	28	13	20	15
Knowledge and technology outputs, rank [2]	62	3	1	15
Institutions, rank [2]	131	16	44	33
R&D spending, % of GDP [3]	1.10 (2020)	3.59 (2022)	2.56 (2022)	2.22 (2021)
Education spending, % of GDP [1]	4.2 (2023)	5.4 (2021)	4.0 (2023)	5.3 (2022)
Real GDP, PPP, $ trillion, 2024 [1]	6.09	25.68	33.60	3.73

Education. Russia's education base is broad: literacy is 99.9% and children can expect 15 years of schooling [1]. Education spending (4.2% of GDP) is similar to China's but lower than in the United States and France. Human capital is the one area where Russia is close to its rivals. These figures measure spending and access, not quality, and the years differ by country.

Technology. Russia spends about 1.1% of GDP on research and development (R&D), about a third of the U.S. level and less than half of China's or France's [3]. On knowledge and technology outputs it ranks 62nd, while China ranks 1st [2]. Its strengths are concentrated in older sectors such as space, where it runs one of the world's largest programs [1]. EU export restrictions on dual-use and advanced technology also cut it off from many foreign parts [4]. The game reflects this history: Russia starts with the most know-how (tacit knowledge, or TK, of 7, against 4 for France and 0 for the United States and China), but it ties France for the smallest budget, 8 dice per round [8].

Intelligence and immigration. Russia has a capable intelligence service; the UK government links major cyber operations to Russian military intelligence (GRU) units such as Unit 26165 [5]. Its main talent problem is skilled workers leaving: about 100,000 IT specialists, roughly 10% of the industry, left in 2022, according to the government's own figures [6]. Overall net migration is still slightly positive (+0.76 per 1,000 people) [1], which hides this loss. The game builds in the same pattern. Russia has the smallest talent pool (K = 5) and loses 40% of its talent each round, but its closed borders mean no other team can steal its research points. The United States and France have open borders and can be robbed [8].

What this means. Russia cannot outspend its rivals at the start, and it cannot hold on to talent. Its advantages are know-how and security. Because the final score multiplies research points by a know-how bonus [8], Russia should first grow its budget, then spend it on the higher-value research that its know-how makes more likely to succeed.

Education recommendation (Tiger)

In the real world, Russia's strength is broad education: near-universal literacy, 15 years of schooling, and a human capital ranking (28th) well above its overall innovation ranking (60th) [1, 2]. Its weakness is keeping and using that talent. Skilled people leave, and weak institutions limit how well the rest are used. The game copies this. Buying education raises our talent pool (K), but every round Russia's talent is multiplied by 0.60, including talent we just bought. One university purchase adds 1 point of talent, which shrinks to 0.6 after one round and 0.1 after four [8].

I recommend that we not spend on education. Talent only matters once it reaches certain levels: 7, to avoid a penalty on medium-level research; 10, for an extra basic research success; and 12, to avoid a penalty on high-risk research [8]. We start at 5. Even if we bought both kinds of education every round, the 40% loss would keep our talent near 3, so we would never reach any of these levels. Those dice do more good if we save them, because each saved die adds 4/3 of a die to every later budget [8].

This is different from what Russia should do in the real world, where protecting a strong education system makes sense. In the game, our talent drains faster than we can replace it. The rulebook asks what a country should do when its talent drains away [8]. Our answer is to rely on know-how instead: every research die we buy adds to our know-how, which improves our odds on harder research and raises our final score. I will watch our talent-flow number each round. It depends partly on wealth, so if our savings push it close to 1.0, education could start to pay off, and I would recommend buying it then.

Technology strategy recommendation (Steve)

Russia's real technology base is narrow but deep. It spends about 1.1% of GDP on R&D, far less than any rival [3], and EU export controls block many advanced parts [4]. The government has responded by focusing state money on areas where Russia is already strong, such as space, nuclear, and defense. The GII still shows Russia producing more than its spending would suggest [2]. The game rewards the same kind of focus. Research comes in three levels: low-risk (L), medium (M), and high-risk (H). Points from medium research count double and points from high-risk research count triple, and Russia's know-how makes the big medium and high-risk rewards more likely [8].

In the first three rounds, I recommend buying only the two cheapest research dice each round, at the low-risk level. The price of research dice rises within a round (1, 1, 2, 2, 3, then 3 dice) and resets the next round, so the first two are the best value in the game. Each one earns about 1.5 points on average and adds 1 to our know-how even if it fails [8]. We should not buy basic research yet. Medium research opens when basic research reaches 8, and high-risk research opens at 20. Other teams' basic research counts 75% toward ours, so their spending moves us toward these levels for free, while our own spending would help them in the same way [8].

From Round 4, once saving has grown our budget to about 70 dice, we should buy enough basic research to open medium and high-risk research, and then buy the most research allowed each round (3 low-risk, 2 medium, and 1 high-risk die). Once our know-how reaches about 25, a medium or high-risk die is worth nearly four times a low-risk die toward our score. Because Russia's talent is low, the game counts our know-how as 10 lower at these levels, so we will keep adding to it with a few extra basic research dice each round [8]. Gilli and Gilli argue that complex technology depends on hands-on experience that cannot simply be copied [7]. The game rewards that kind of knowledge, and Russia starts with the most of it.

Intelligence and immigration recommendation (Gloria)

In the real world, Russia has strong intelligence capabilities but struggles to keep talent. The UK government links major cyber operations to Russian military intelligence [5], while about 100,000 IT specialists left the country in 2022 [6]. Weak institutions (131st) make it hard to attract skilled workers or win them back [2]. Russia's actual policy favors tight research security, and the game starts Russia the same way, with closed borders [8].

I recommend keeping our borders closed for the whole game. Opening them would not bring us more talent: the rulebook shows that Russia opening changes no country's talent flow, because talent goes to rich countries with strong universities [8]. Opening would only let other teams steal our research points, with a one-in-three chance for each spy. While we stay closed, no one can steal from us. Openness pays off for a rich country with top universities like the United States; for Russia it is all risk.

For spying, I recommend no spies in Round 1 and two spies per round after that, at 1 die each. A spy can copy a rival's points at one research level, but in Round 1 no one has any points yet. Only countries with open borders can be robbed. Against the United States or France, a spy steals one time in three, and a spy on low-risk research is never caught. Spying on medium or high-risk research is riskier (caught one time in six and one time in three), but one successful steal there is worth far more [8]. We will start at the low-risk level in Round 2 and move to the higher levels once a rival's public basic research shows it can reach them. We will not spy on China: its borders are closed, so we could only look, never steal, and we would be caught more often.

Executive recommendation (Steve)

Decision rule. Each czar proposes the plan for their own area, and all three of us vote. Two votes decide. As executive, I break ties and make the final call if the round timer (20 minutes in Round 1, 10 minutes after that) is about to run out. All three of us share note-taking during the game, as the instructor asked.

Overall strategy. Our strategy is "save, then strike." Two-thirds of every unspent die becomes wealth at the start of the next round, and each round's budget is twice our wealth [8], so saving about three-quarters of the budget roughly doubles it every round. The game runs over three 80-minute classes with 10-minute rounds, and the game portal shows a 10-round limit, so we expect 8 to 10 rounds, and the rounds in the first class are very unlikely to be the last. In our own simulation of the rules for 8 to 10 rounds, this plan scored about 30% higher on average than spending everything each round. The table below shows the plan round by round.

Round	Budget (dice)	What we buy	What we save	Why
1	8	2 low-risk research dice	6	Cheapest research; nothing to steal yet
2 and 3	16, then about 32	2 low-risk research dice; 2 spies on the US and France at the low-risk level	About three-quarters	Grow the budget while saving is safe
4 onward (second class)	About 70	Basic research to open medium (8) and high-risk (20) research; then 3 low-risk, 2 medium, and 1 high-risk die each round; spies at higher levels	Only what we cannot use well	Turn wealth into high-value points
Third class	Varies	Almost everything	Little or nothing	Savings in the final round are lost

In every round we keep our borders closed and buy no education.

Risks, changes, and agreement. The last round is not announced, and anything saved in it is lost [8], so after Round 3 we will save only dice we cannot put to good use. Our growing wealth is public, so rivals may copy us or see us as a threat, but because our borders are closed they cannot steal from us. We will start spending earlier if other teams' basic research opens medium research quickly or if the instructor signals a shorter game. We reviewed every recommendation together, and all three members agreed with each one, so the plan was adopted unanimously and the tiebreaker was not needed.

References

[1] Central Intelligence Agency. *The World Factbook*, entries for Russia, United States, China, and France (archived data, reproduced at theworldfactbook.org after the CIA retired the publication in February 2026). Accessed November 2026.

[2] World Intellectual Property Organization. *Global Innovation Index 2025* country profiles: Russian Federation, United States, China, France.

[3] National Science Board. *Science and Engineering Indicators 2025* (NSB-2025-7), Table DISC-10: GERD, GDP, and GERD-to-GDP ratio by selected country. Russia's latest reported year is 2020.

[4] European Commission. *Export-related restrictions for dual-use goods and advanced technologies*, FAQ on Council Regulation (EU) No 833/2014, updated September 25, 2026.

[5] UK Government. *Profile: GRU cyber and hybrid threat operations*, updated July 13, 2026.

[6] Interfax. "About 100,000 IT specialists left Russia in 2022 - digital development minister," December 20, 2022.

[7] Gilli, Andrea, and Mauro Gilli. 2019. "Why China Has Not Caught Up Yet: Military-Technological Superiority and the Limits of Imitation, Reverse Engineering, and Cyber Espionage." *International Security* 43(3): 141-189. doi:10.1162/isec_a_00337.

[8] Canfil, Justin K. *Global Technology Competition: Official Rulebook*, executive briefing video, and player guide, course materials, November 2026.